

HOUSEHOLD AI / FINDINGS BRIEF

# Your home does not need one giant AI.

A findings brief on matching household jobs to the right capability, device, data boundary, and level of authority.

## The core idea

A household should use a measured portfolio of trusted capabilities, not send every request to one general model.

Research synthesis and design hypothesis. This brief does not claim an implemented runtime or original empirical validation.

# Household AI begins with the job, not the model.

A bill check, a homework question, a grocery plan, and a safety alert do not need the same intelligence, information, hardware, or authority. Household surveys show recurring clusters of administrative, care, learning, shopping, maintenance, transportation, and leisure activity; they are demand evidence, not an automation mandate. [1][2]

The design question is therefore not simply which model is smartest. It is which option has evidence that it can finish this task, where it can run within privacy and resource limits, what it may do, and how it hands off when the task exceeds its scope.

|                                      |                                                                                                                                                                          |
|--------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. A home is not one workload.       | Household jobs differ in urgency, privacy, stakes, frequency, and who may act. A generic chat surface hides those differences.                                           |
| 2. Bigger is not the right default.  | The right option may be a rule, a small specialist, a general model, a remote service, or a person. Use the least demanding option shown to meet the bar, then escalate. |
| 3. Capable does not mean authorized. | A device may have enough compute and still be the wrong place for financial, child, guest, or employer-managed information.                                              |

# The durable product sits above replaceable models.

|                                              |                                                                                                                                             |
|----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| 4. The model can change. The promise cannot. | Outcome, information scope, authority, approval, recovery, and user experience should remain stable as model and hardware suppliers change. |
| 5. The lasting value sits above the model.   | Trusted knowledge of household jobs, permissions, corrections, and successful outcomes compounds. Model access alone does not.              |

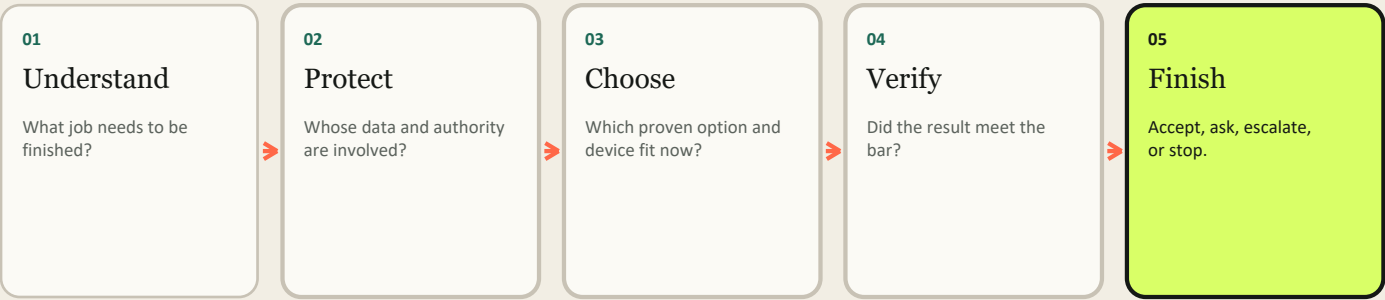
## A small reference task profile

| Household job            | What changes                               | Default boundary                       |
|--------------------------|--------------------------------------------|----------------------------------------|
| Bills and administration | Evidence, account authority, reversibility | Read and draft; approval before action |
| Learning support         | Age, subject scope, assessed work          | Teach; do not impersonate or complete  |
| Maintenance              | Exact device, manual, physical safety      | Low-risk guidance; stop for hazards    |
| Shared shopping          | Inventory, allergies, private preferences  | Combine only explicitly shared context |

The point is not to hard-code four assistants. It is to make task evidence, information scope, authority, and stopping behavior explicit before choosing a model or device.

# One request. Five plain-language decisions.

The household does not choose a model, processor, or cloud. It states the job. The system checks the boundaries, selects an eligible option, validates the result, and either finishes or hands off.



**Fallback keeps the same rules.**

If the first option cannot finish, the system may try another approved device, minimize context for a remote model, ask a person, defer, or stop. It never weakens the original privacy, quality, or authority limits.

## What is being coordinated

|                                                                                                                                 |                                                                                                                                       |
|---------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| <p>Proven capability</p> <p>A model, rule, knowledge pack, tool, or validator with a measured competence boundary.</p>          | <p>Allowed device</p> <p>A phone, PC, hub, appliance, or remote service that fits memory, runtime, energy, and trust constraints.</p> |
| <p>Scoped context</p> <p>Personal, shared, child, guest, and employer information remain separate unless explicitly shared.</p> | <p>Guarded outcome</p> <p>Accept, ask permission, retry, undo, hand off, defer, or deny - with evidence of what happened.</p>         |

# Useful because the stopping points are clear.

|                                                                                                                                                                                                                                   |                                                                                                                                                                                                                 |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>Catch a bill increase</b></p> <p>Compare the new statement with prior charges and cite the evidence behind the change.</p> <p>Boundary: Draft a question, but do not pay, cancel, or close an account without approval.</p> | <p><b>Help a child learn</b></p> <p>Use assigned material to explain the next step and check understanding.</p> <p>Boundary: Do not complete assessed work; hand off when outside the subject or age range.</p> |
| <p><b>Troubleshoot an appliance</b></p> <p>Find the exact model and manual before suggesting bounded checks.</p> <p>Boundary: Stop for electrical, gas, structural, hazardous, or warranty-voiding work.</p>                      | <p><b>Plan groceries</b></p> <p>Combine authorized lists, inventory, allergies, and shared preferences.</p> <p>Boundary: Do not use personal information that was never explicitly shared.</p>                  |

These examples are not claims of empirical performance. They are design tests: can the product make its useful outcome, evidence, authority, and stopping point legible to a household member?

# The control point moves above the model.

| Builders                                                                                                                                                       | Device makers                                                                                                                                                              | Investors                                                                                                                                                                  |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Start with the household job.<br>Define the useful result, minimum information, forbidden actions, approval point, and correction path before binding a model. | Prove useful work, not peak AI.<br>Publish measured task capability under real memory, battery, heat, privacy, and foreground-use constraints - not only accelerator TOPS. | Model access is not the moat.<br>Look for trusted outcomes, permission relationships, task evidence, cross-device reach, supplier substitutability, and recovery behavior. |

## The economic rule

Optimize for cost per policy-compliant successful outcome, not token price. Count depreciation, energy, storage, cold starts, device interference, update traffic, validation, routing mistakes, human review, and failure loss. Routing research already shows that cost and quality can improve through selective use of different models; the household question adds device and authority constraints. [7][8][9]

|                               |                                                                                                                                                                                                                                                                                        |
|-------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| The honest falsification test | This thesis loses if one compact general model plus tools delivers the same task success, privacy, latency, authority control, and full cost with less complexity. It also loses if specialist competence cannot be detected reliably or portfolio maintenance overwhelms the savings. |
|-------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

# The ingredients exist. The household framing is the proposal.

Fog, edge, and service-routing standards already describe placement across heterogeneous infrastructure. Model routers already trade quality and price. On-device frameworks, smart-home pipelines, and inference serving systems already select runtimes and implementations. Privacy systems already use attestation and minimized cloud processing. [3][4][5][6][7][8][9][10][11][12][13]

The defensible hypothesis is narrower: a portable household product layer that joins recurring household jobs, measured specialist competence, device feasibility, scoped identity and knowledge, explicit authority, validation, and safe fallback. This is a synthesis to test, not a claim that hybrid or edge inference was invented here.

## Non-negotiable boundaries

|                                                                                                           |                                                                                                                   |
|-----------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| Competence before placement<br>A device is considered only after the capability is shown to fit the task. | Policy before optimization<br>A cheaper or faster route cannot weaken privacy, residency, identity, or authority. |
| Abstention is a result<br>A system that cannot meet the bar must ask, hand off, defer, or stop. [14]      | Feedback does not rewrite policy<br>Observed outcomes can propose changes; deliberate governance approves them.   |

Bottom line: The future of household AI is unlikely to be one permanent model on one permanent device. The opportunity is to make a changing portfolio feel like one stable, trustworthy household capability.

# Evidence behind the brief

Sources prioritize official standards, first-party technical documentation, and peer-reviewed or clearly labeled preprint research. Accessed July 2026.

- [1] U.S. Bureau of Labor Statistics. American Time Use Survey Overview. <https://www.bls.gov/tus/overview.htm>
- [2] U.S. Bureau of Labor Statistics. Consumer Expenditure Surveys. <https://www.bls.gov/cex/>
- [3] NIST. SP 500-325: Fog Computing Conceptual Model. <https://csrc.nist.gov/pubs/sp/500/325/final>
- [4] IEEE. 1934-2018: OpenFog Reference Architecture. <https://standards.ieee.org/ieee/1934/7137/>
- [5] ETSI. MEC 003 v4.1.1: Framework and Reference Architecture. [https://www.etsi.org/deliver/etsi\\_gs/mec/001\\_099/003/04.01.01\\_60/gs\\_mec003v040101p.pdf](https://www.etsi.org/deliver/etsi_gs/mec/001_099/003/04.01.01_60/gs_mec003v040101p.pdf)
- [6] IETF. Computing-Aware Traffic Steering (CATS). <https://datatracker.ietf.org/group/cats/about/>
- [7] Lingjiao Chen, Matei Zaharia, James Zou. FrugalGPT (preprint, 2023). <https://arxiv.org/abs/2305.05176>
- [8] LMSYS Org. RouteLLM (preprint, 2024). <https://arxiv.org/abs/2406.18665>
- [9] Francisco Romero et al. INFaaS (preprint, 2019). <https://arxiv.org/abs/1905.13348>
- [10] Apple. Private Cloud Compute Security Guide. <https://security.apple.com/documentation/private-cloud-compute/>
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- [13] Microsoft Learn. Windows ML overview. <https://learn.microsoft.com/en-us/windows/ai/new-windows-ml/overview>
- [14] Yonatan Geifman and Ran El-Yaniv. Selective Classification for Deep Neural Networks. NeurIPS 2017. <https://papers.nips.cc/paper/2017/hash/4a8423d5e91fda00bb7e46540e2b0cf1-Abstract.html>

Scope note: household task surveys show how people spend time and money; they do not prove that a task should be automated. Performance values, routing gains, and economics require future empirical validation.